

DBMS_Lab_Manual_SQL_Joins_Transactions:

Roll No:2520030605

```
Microsoft Windows [Version 10.0.26200.8894]
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C:\Users\hp>cd "C:\Program Files\MySQL\MySQL Server 8.0\bin"

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 18
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE DBMS_LAB;
Query OK, 1 row affected (0.05 sec)

mysql> USE DBMS_LAB;
Database changed
mysql> CREATE TABLE roles(
    -> role_id INT PRIMARY KEY,
    -> role_name VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.23 sec)

mysql> CREATE TABLE users(
    -> user_id INT PRIMARY KEY,
    -> name VARCHAR(50),
    -> email VARCHAR(50),
    -> role_id INT
    -> );
Query OK, 0 rows affected (0.11 sec)
```

```
mysql> CREATE TABLE departments(  
    -> dept_id INT PRIMARY KEY,  
    -> dept_name VARCHAR(50)  
    -> );
```

Query OK, 0 rows affected (0.15 sec)

```
mysql> CREATE TABLE employees(  
    -> emp_id INT PRIMARY KEY,  
    -> emp_name VARCHAR(50),  
    -> dept_id INT,  
    -> manager_id INT,  
    -> salary DECIMAL(10,2)  
    -> );
```

Query OK, 0 rows affected (0.05 sec)

```
mysql> CREATE TABLE customers(  
    -> customer_id INT PRIMARY KEY,  
    -> customer_name VARCHAR(50),  
    -> city VARCHAR(50)  
    -> );
```

Query OK, 0 rows affected (0.05 sec)

```
mysql> CREATE TABLE orders(  
    -> order_id INT PRIMARY KEY,  
    -> customer_id INT,  
    -> order_date DATE,  
    -> total_amount DECIMAL(10,2)  
    -> );
```

Query OK, 0 rows affected (0.09 sec)

```
mysql> CREATE TABLE accounts(  
    -> account_no INT PRIMARY KEY,  
    -> account_no INT PRIMARY KEY,  
    -> balance DECIMAL(10,2)  
    -> );
```

ERROR 1060 (42S21): Duplicate column name 'account_no'

```
mysql> CREATE TABLE accounts(  
-> account_no INT PRIMARY KEY,  
-> holder_name VARCHAR(50),  
-> balance DECIMAL(10,2)  
-> );  
Query OK, 0 rows affected (0.10 sec)  
  
mysql> INSERT INTO roles VALUES  
-> (1,'Admin'),  
-> (2,'Manager'),  
-> (3,'Employee'),  
-> (4,'HR');  
Query OK, 4 rows affected (0.08 sec)  
Records: 4 Duplicates: 0 Warnings: 0  
  
mysql> INSERT INTO users VALUES  
-> (101,'Ramesh','ramesh@gmail.com',1),  
-> (102,'Suresh','suresh@gmail.com',2),  
-> (103,'Priya','priya@gmail.com',3),  
-> (104,'Kiran','kiran@gmail.com',NULL);  
Query OK, 4 rows affected (0.07 sec)  
Records: 4 Duplicates: 0 Warnings: 0  
  
mysql> INSERT INTO departments VALUES  
-> (1,'CSE'),  
-> (2,'ECE'),  
-> (3,'MECH');  
Query OK, 3 rows affected (0.01 sec)  
Records: 3 Duplicates: 0 Warnings: 0  
  
mysql> INSERT INTO employees VALUES  
-> (1,'Anil',1,NULL,80000),  
-> (2,'Sunil',1,1,60000),  
-> (3,'Rahul',2,1,45000),  
-> (4,'Meena',3,2,55000),  
-> (5,'Divya',1,2,70000);  
Query OK, 5 rows affected (0.08 sec)  
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO customers VALUES
-> (1,'Arun','Hyderabad'),
-> (2,'Bhanu','Vijayawada'),
-> (3,'Charan','Warangal');
Query OK, 3 rows affected (0.08 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO orders VALUES
-> (1001,1,'2026-06-01',15000),
-> (1002,2,'2026-06-02',20000),
-> (1003,1,'2026-06-05',10000),
-> (1004,3,'2026-06-07',25000);
Query OK, 4 rows affected (0.08 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO accounts VALUES
-> (101,'Ravi',50000),
-> (102,'Kumar',30000),
-> (103,'Sita',45000);
Query OK, 3 rows affected (0.08 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT emp_name,dept_name
-> FROM employees e
-> INNER JOIN departments d
-> ON e.dept_id=d.dept_id;
```

```
+-----+-----+
| emp_name | dept_name |
+-----+-----+
| Anil     | CSE       |
| Sunil    | CSE       |
| Rahul    | ECE       |
| Meena    | MECH      |
| Divya    | CSE       |
+-----+-----+
5 rows in set (0.08 sec)
```

```
mysql> SELECT name,email,role_name
-> FROM users u
-> LEFT JOIN roles r
-> ON u.role_id=r.role_id;
```

name	email	role_name
Ramesh	ramesh@gmail.com	Admin
Suresh	suresh@gmail.com	Manager
Priya	priya@gmail.com	Employee
Kiran	kiran@gmail.com	NULL

4 rows in set (0.07 sec)

```
mysql> SELECT name,role_name
-> FROM users u
-> RIGHT JOIN roles r
-> ON u.role_id=r.role_id;
```

name	role_name
Ramesh	Admin
Suresh	Manager
Priya	Employee
NULL	HR

4 rows in set (0.00 sec)

```
mysql> SELECT name,email,role_name
-> FROM users u
-> LEFT JOIN roles r
-> ON u.role_id=r.role_id
-> UNION
-> SELECT name,email,role_name
-> FROM users u
-> RIGHT JOIN roles r
-> ON u.role_id=r.role_id;
```

name	email	role_name
Ramesh	ramesh@gmail.com	Admin
Suresh	suresh@gmail.com	Manager
Priya	priya@gmail.com	Employee
Kiran	kiran@gmail.com	NULL
NULL	NULL	HR

5 rows in set (0.01 sec)

```
mysql> SELECT e.emp_name AS Employee,
-> m.emp_name AS Manager
-> FROM employees e
-> LEFT JOIN employees m
-> ON e.manager_id=m.emp_id;
```

Employee	Manager
Anil	NULL
Sunil	Anil
Rahul	Anil
Meena	Sunil
Divya	Sunil

5 rows in set (0.05 sec)

```
mysql> SELECT o.order_id,
-> customer_name,
-> city,
-> total_amount
-> FROM customers c
-> JOIN orders o
-> ON c.customer_id=o.customer_id;\
```

order_id	customer_name	city	total_amount
1001	Arun	Hyderabad	15000.00
1002	Bhanu	Vijayawada	20000.00
1003	Arun	Hyderabad	10000.00
1004	Charan	Warangal	25000.00

4 rows in set (0.00 sec)

```
mysql> SELECT dept_name,
-> COUNT(*) AS EmployeeCount,
-> AVG(salary) AS AverageSalary,
-> MAX(salary) AS MaximumSalary
-> FROM employees e
-> JOIN departments d
-> ON e.dept_id=d.dept_id
-> GROUP BY dept_name;
```

dept_name	EmployeeCount	AverageSalary	MaximumSalary
CSE	3	70000.000000	80000.00
ECE	1	45000.000000	45000.00
MECH	1	55000.000000	55000.00

3 rows in set (0.01 sec)

```
mysql> SELECT emp_name,salary,dept_name
-> FROM employees e
-> JOIN departments d
-> ON e.dept_id=d.dept_id
-> WHERE salary>50000
-> AND dept_name='CSE';
```

emp_name	salary	dept_name
Anil	80000.00	CSE
Sunil	60000.00	CSE
Divya	70000.00	CSE

3 rows in set (0.05 sec)

mysql> ^C

```
mysql> SELECT customer_name,
      -> SUM(total_amount) AS TotalPurchase
      -> FROM customers c
      -> JOIN orders o
      -> ON c.customer_id=o.customer_id
      -> GROUP BY customer_name
      -> ORDER BY TotalPurchase DESC;
```

customer_name	TotalPurchase
Arun	25000.00
Charan	25000.00
Bhanu	20000.00

3 rows in set (0.05 sec)

```
mysql> CREATE TABLE faculty(
      -> faculty_id INT PRIMARY KEY,
      -> faculty_name VARCHAR(50)
      -> );
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> CREATE TABLE courses(
      -> course_id INT PRIMARY KEY,
      -> course_name VARCHAR(50),
      -> faculty_id INT
      -> );
Query OK, 0 rows affected (0.07 sec)
```

```
mysql> CREATE TABLE students(  
  -> student_id INT PRIMARY KEY,  
  -> student_name VARCHAR(50)  
  -> );  
Query OK, 0 rows affected (0.04 sec)  
  
mysql> CREATE TABLE enrollments(  
  -> enrollment_id INT PRIMARY KEY,  
  -> student_id INT,  
  -> course_id INT  
  -> );  
Query OK, 0 rows affected (0.12 sec)  
  
mysql> INSERT INTO faculty VALUES  
  -> (1, 'Dr Rao'),  
  -> (2, 'Dr Kumar');  
Query OK, 2 rows affected (0.06 sec)  
Records: 2  Duplicates: 0  Warnings: 0  
  
mysql> INSERT INTO courses VALUES  
  -> (101, 'DBMS', 1),  
  -> (102, 'Java', 2);  
Query OK, 2 rows affected (0.02 sec)  
Records: 2  Duplicates: 0  Warnings: 0  
  
mysql> INSERT INTO students VALUES  
  -> (1, 'Ganesh'),  
  -> (2, 'Sai');  
Query OK, 2 rows affected (0.08 sec)  
Records: 2  Duplicates: 0  Warnings: 0  
  
mysql> INSERT INTO enrollments VALUES  
  -> (1, 1, 101),  
  -> (2, 2, 102);  
Query OK, 2 rows affected (0.04 sec)  
Records: 2  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT student_name,
-> course_name,
-> faculty_name
-> FROM students s
-> JOIN enrollments e
-> ON s.student_id=e.student_id
-> JOIN courses c
-> ON e.course_id=c.course_id
-> JOIN faculty f
-> ON c.faculty_id=f.faculty_id;
+-----+-----+-----+
| student_name | course_name | faculty_name |
+-----+-----+-----+
| Ganesh       | DBMS        | Dr Rao       |
| Sai          | Java        | Dr Kumar     |
+-----+-----+-----+
2 rows in set (0.05 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> INSERT INTO accounts
-> VALUES(104,'Anil',25000);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> COMMIT;
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> INSERT INTO accounts
-> VALUES(105,'Ravi',10000);
Query OK, 1 row affected (0.04 sec)
```

```
mysql> ROLLBACK;
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE accounts
  -> SET balance=balance-5000
  -> WHERE account_no=101;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE accounts
  -> SET balance=balance+5000
  -> WHERE account_no=102;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> COMMIT;
Query OK, 0 rows affected (0.05 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE accounts
  -> SET balance=balance-3000
  -> WHERE account_no=101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE accounts
  -> SET balance=balance+3000
  -> WHERE account_no=999;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql> ROLLBACK;
Query OK, 0 rows affected (0.06 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> UPDATE accounts  
      -> SET balance=balance+2000  
      -> WHERE account_no=101;  
Query OK, 1 row affected (0.00 sec)  
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> SAVEPOINT s1;  
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> UPDATE accounts  
      -> SET balance=balance-1000  
      -> WHERE account_no=101;  
Query OK, 1 row affected (0.00 sec)  
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> ROLLBACK TO s1;  
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> COMMIT;  
Query OK, 0 rows affected (0.08 sec)
```